SDLC, Process Models and Requirements

Software Engineering - Units 1-2 - Building software systematically

Software Development Life Cycle

The SDLC organizes work into phases: requirements, design, implementation, testing, deployment and maintenance. Process models arrange these phases differently.

Process models

- Waterfall: sequential phases; simple and well documented but inflexible when requirements change.
- Incremental: deliver the product in working slices.
- Spiral: iterative with explicit risk analysis in each loop; suited to large, risky projects.
- Agile (Scrum, Kanban): short iterations, working software, continuous feedback, close customer collaboration.

Scrum at a glance

- Roles: Product Owner, Scrum Master, Development Team.
- Artifacts: product backlog, sprint backlog, increment.
- Events: sprint planning, daily stand-up, sprint review, retrospective.

Requirements engineering

Functional requirements say what the system does (a student can download a PDF). Non-functional requirements say how well it does it (pages load in under two seconds, works on phones). Good requirements are testable, unambiguous and traceable.

> *User-story template: As a <role>, I want <goal> so that <benefit>. Add acceptance criteria that can be checked yes/no.*

UML diagrams you should know

- Use case diagram - actors and what they can do.
- Class diagram - classes, attributes, relationships.
- Sequence diagram - message order between objects over time.
- Activity and state diagrams - workflows and life cycles.